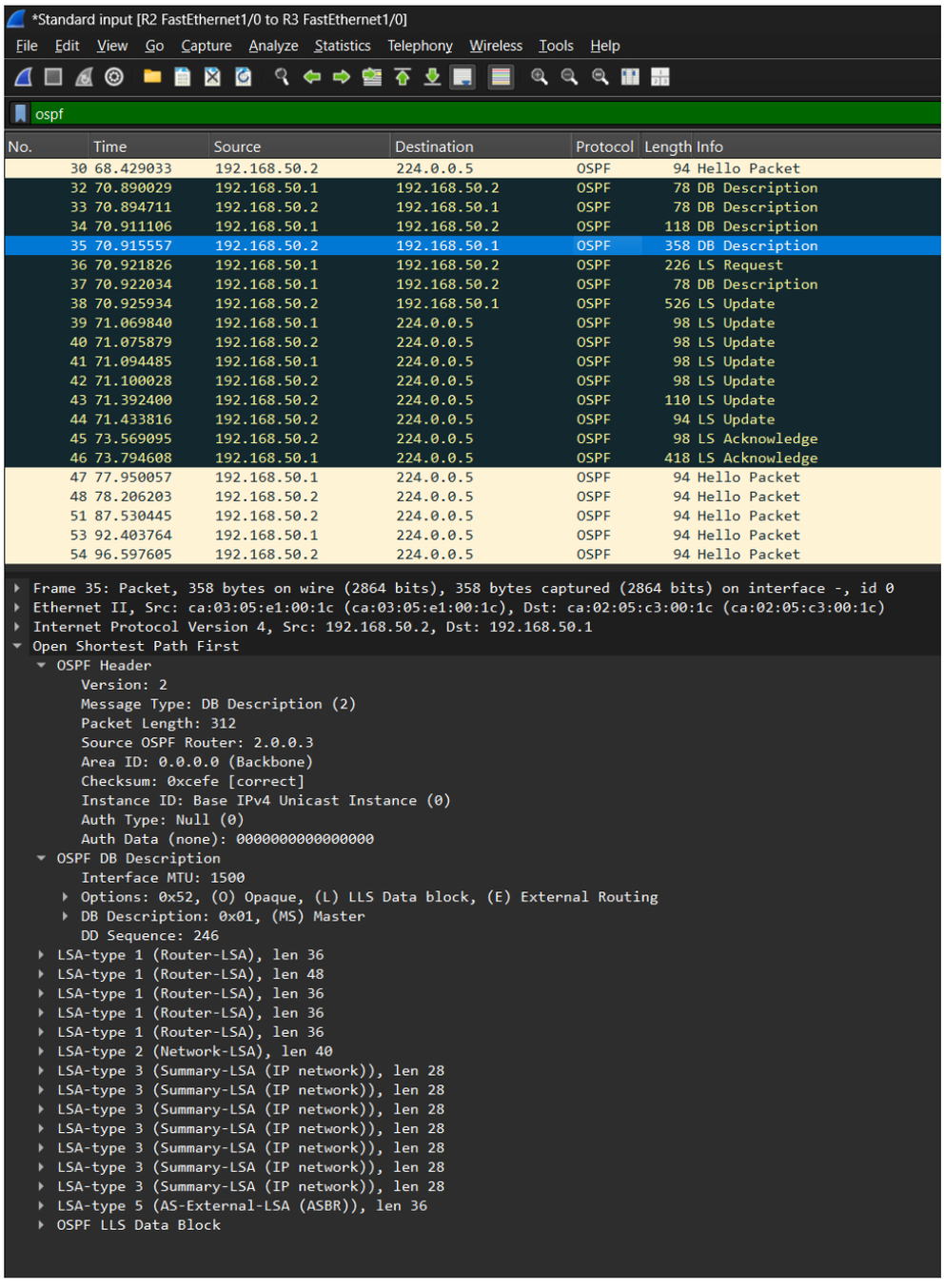
1.

a.

The DB Description packets are used to synchronize the LSDB between neighboring routers. These packets do not carry full LSAs, but only their headers, allowing each router to compare its database with the neighbor's.

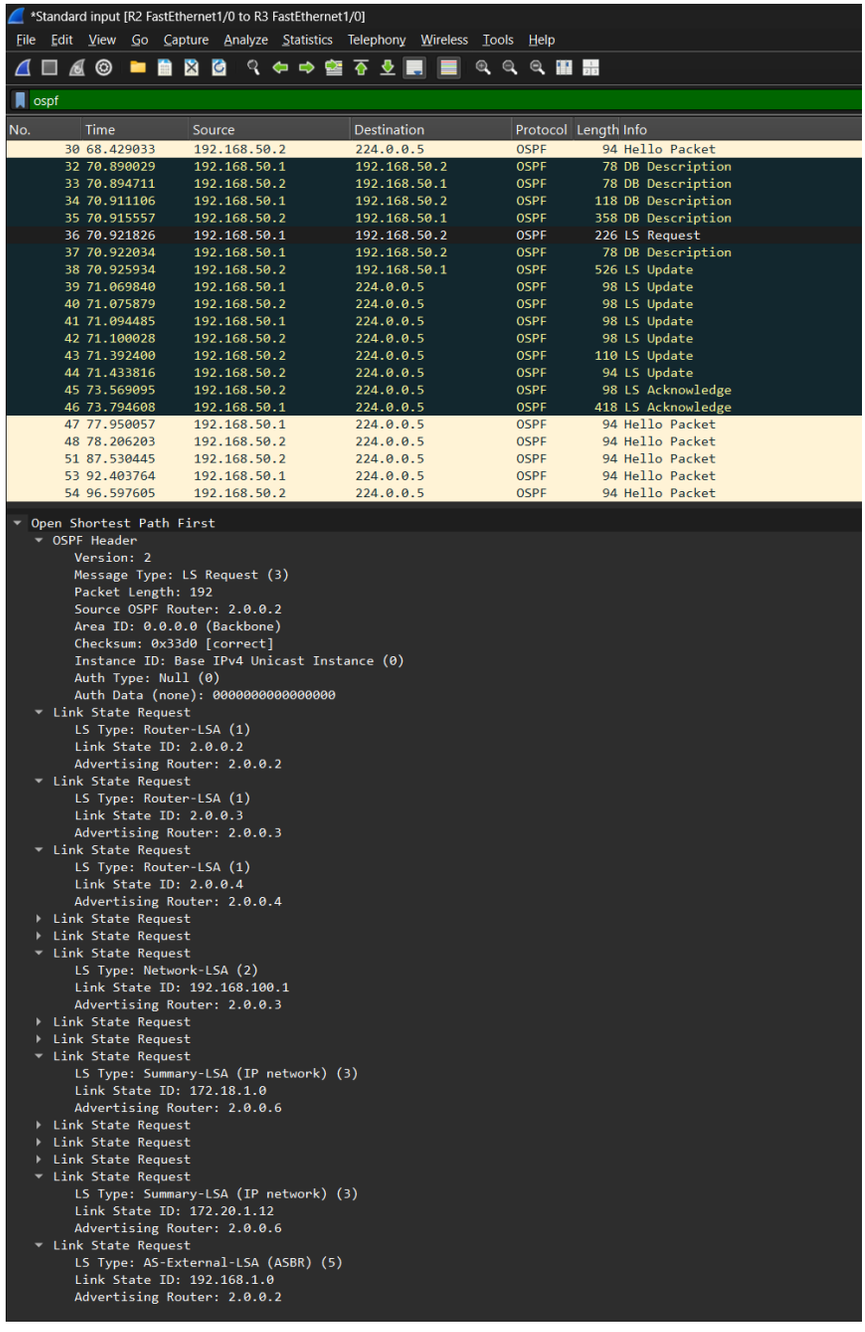
From this exchange, each router identifies which LSAs are missing or outdated and later requests them using LS Request messages. This process is required whenever a new adjacency is established, since routers must ensure that their LSDBs are identical in order to have a consistent view of the network topology



b.

LS Request packets are used to request specific LSAs that are missing or outdated in a router's LSDB. Each request includes identifiers such as the LSA Type, Link State ID, and Advertising Router, which uniquely identify the required information.

A router determines what to request by comparing the LSA headers received in DB Description packets with its own LSDB. Any discrepancies detected in this comparison lead to the generation of LS Request messages for the missing or outdated LSAs.



2.

a.

R6#show ip ospf database router 2.0.0.6

In area 0, R6 advertises the multi-access network 192.168.100.0/24 as a transit network, with interface address 192.168.100.2. In area 1, R6 advertises the link towards R7 as a transit network, with Designated Router address 172.20.1.2 and local interface address 172.20.1.1. In area 2, R6 advertises a point-to-point link to router R8, identified by neighbor router ID 2.0.0.8 and local interface address 172.20.1.14, and also advertises the corresponding stub network 172.20.1.12/30.

b.

In area 1, R6 advertises the link as a transit network. This happens because the connection is made through a FastEthernet interface, which OSPF treats as a broadcast multi-access network. As a result, the link is represented through the Designated Router address and the local interface address.

In area 2, the link is advertised differently because R6 is connected to R8 through a serial point-to-point link. In this case, the Router LSA shows a point-to-point connection to the neighboring router R8 and also includes the corresponding stub network 172.20.1.12/30. Therefore, the difference between the advertisements in areas 1 and 2 is directly related to the type of interface used by R6 in each area: broadcast network in area 1 and point-to-point network in area 2.

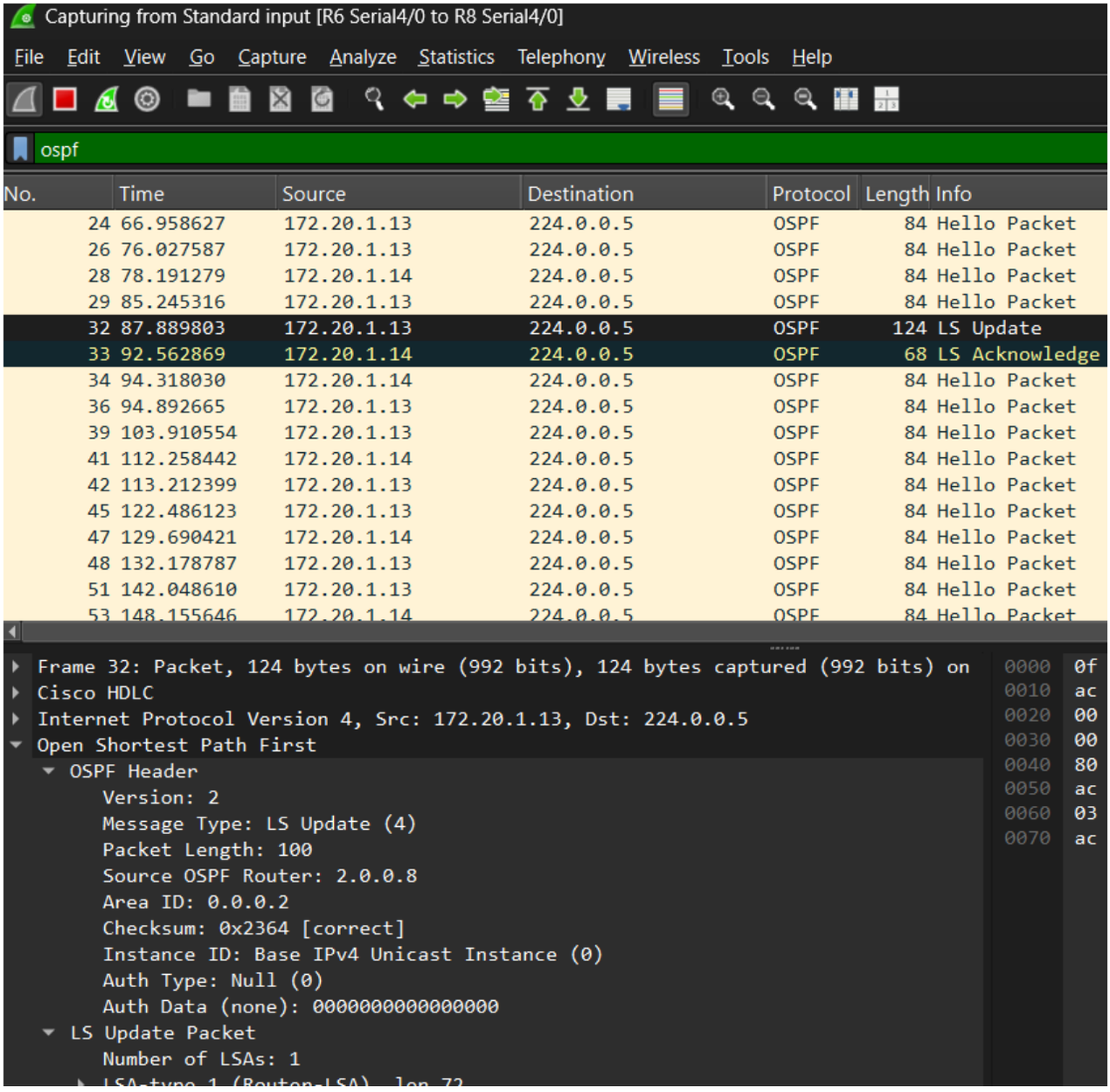
3.

a.

When router R9 was stopped, R8 detected that the neighbor was no longer reachable and the adjacency state changed from FULL to DOWN. This occurred because R8 stopped receiving Hello packets from R9, leading to the expiration of the Dead Timer.

This event triggered a neighbor down condition, as shown in the debug output ("Neighbor Down: Dead timer expired"), causing OSPF to update its neighbor state and recompute the topology.

```
R8#debug ip ospf adj
OSPF adjacency debugging is on
R8#
*Apr  6 14:37:40.107: OSPF-2 ADJ   Fa0/0: 2.0.0.9 address 172.20.1.9 is dead
*Apr  6 14:37:40.107: OSPF-2 ADJ   Fa0/0: 2.0.0.9 address 172.20.1.9 is dead, state DOWN
*Apr  6 14:37:40.107: %OSPF-5-ADJCHG: Process 2, Nbr 2.0.0.9 on FastEthernet0/0 from FULL to DOWN, Neighbor Down: Dead ti
mer expired
*Apr  6 14:37:40.111: OSPF-2 ADJ   Fa0/0: Neighbor change event
*Apr  6 14:37:40.111: OSPF-2 ADJ   Fa0/0: DR/BDR election
*Apr  6 14:37:40.111: OSPF-2 ADJ   Fa0/0: Elect BDR 2.0.0.8
*Apr  6 14:37:40.115: OSPF-2 ADJ   Fa0/0: Elect DR 2.0.0.8
*Apr  6 14:37:40.115: OSPF-2 ADJ   Fa0/0: Elect BDR 0.0.0.0
*Apr  6 14:37:40.115: OSPF-2 ADJ   Fa0/0: Elect DR 2.0.0.8
*Apr  6 14:37:40.119: OSPF-2 ADJ   Fa0/0: DR: 2.0.0.8 (Id)   BDR: none
*Apr  6 14:37:40.119: OSPF-2 ADJ   Fa0/0: Remember old DR 2.0.0.9 (id)
R8#
```



b.

After the failure of R9, R8 detected a topology change in area 2 and generated an **LS Update packet** addressed to 224.0.0.5. This packet contains a single **Router LSA (Type 1)** originated by R8 itself (Advertising Router: 2.0.0.8).

The Router LSA describes the current state of R8's links in the area. In this case, it reflects the updated topology after the loss of the adjacency with R9. The LSA is flooded to all routers in the area so that they can update their LSDB and recompute their shortest path tree.

From the capture, it is also possible to observe that the LSA has a **new sequence number**, indicating that it is a more recent version of the advertisement, and therefore must replace the previous one in the LSDB.

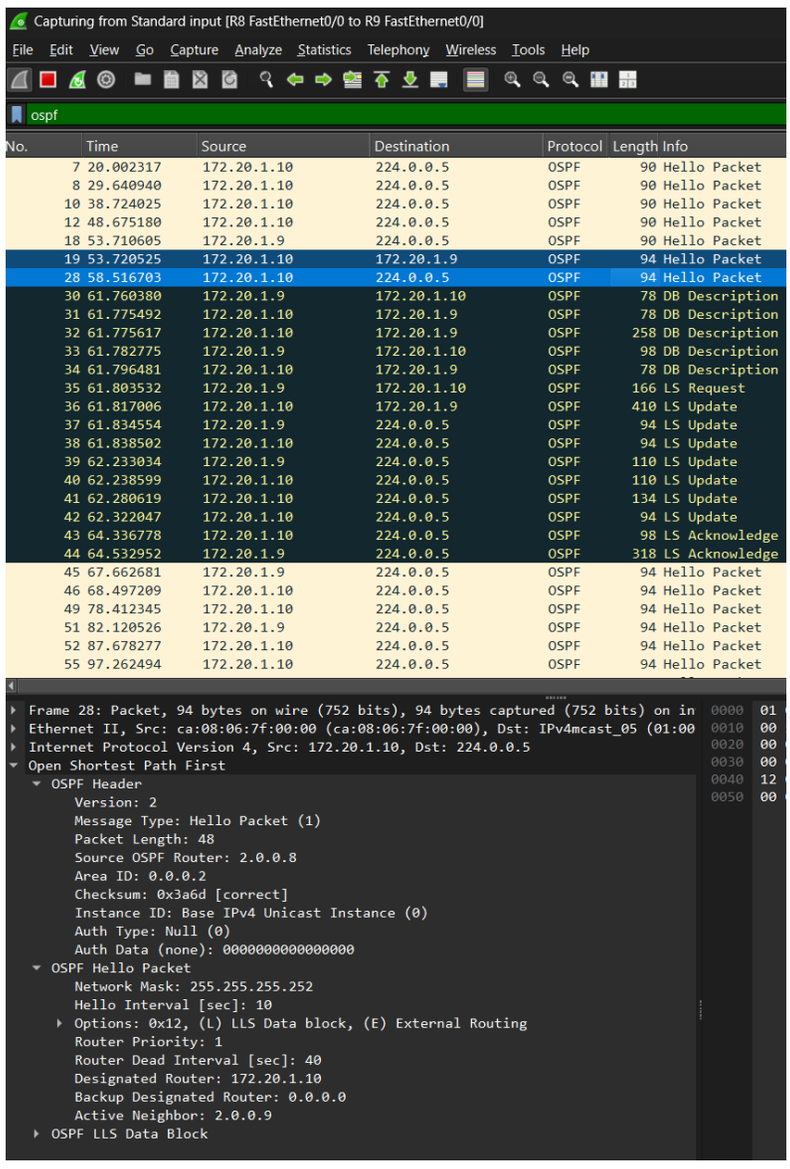
After receiving the LS Update, R6 replies with an **LS Acknowledge**, confirming that the LSA was successfully received. This acknowledgement ensures reliable flooding of LSAs across the network.

c.

The message that allows R9 to know that it has bidirectional communication with R8 is an OSPF Hello packet sent by R8 that includes R9 in the neighbor list.

In the captured packet, R8 (Router ID 2.0.0.8) sends a Hello message where the field Active Neighbor contains the Router ID of R9 (2.0.0.9). This indicates that R8 has already received Hello packets from R9 and recognizes it as a neighbor.

In OSPF, bidirectional communication is confirmed when a router receives a Hello packet from a neighbor that explicitly lists its own Router ID. Therefore, this Hello packet confirms that both routers are receiving each other's messages and can proceed to establish a full adjacency.



d.

During the OSPF convergence process between R8 and R9, several types of LSAs were observed inside the LS Update messages.

The Router LSA (Type 1) was exchanged to describe the local links of each router within the area. These LSAs are fundamental to building the topology graph, as they represent the connections between routers and networks.

The Network LSA (Type 2) was also observed and is generated by the Designated Router in multi-access networks. Its purpose is to describe the set of routers connected to a broadcast network, allowing the topology to be represented efficiently.

Additionally, Summary LSAs (Type 3) were exchanged. These LSAs are generated by Area Border Routers and are used to advertise networks from one area into another, enabling inter-area routing.

Finally, AS-External LSAs (Type 5) were present, which are used to advertise routes external to the OSPF domain. These LSAs allow routers to reach networks learned from outside the autonomous system.

Together, these LSAs ensure that all routers build a consistent and complete view of the network topology, allowing correct path computation using the SPF algorithm.

```
▸ Frame 36: Packet, 410 bytes on wire (3280 bits), 410 bytes captured (3280 bits) o
▸ Ethernet II, Src: ca:08:06:7f:00:00 (ca:08:06:7f:00:00), Dst: ca:09:06:9f:00:00 (
▸ Internet Protocol Version 4, Src: 172.20.1.10, Dst: 172.20.1.9
▾ Open Shortest Path First
  ▾ OSPF Header
      Version: 2
      Message Type: LS Update (4)
      Packet Length: 376
      Source OSPF Router: 2.0.0.8
      Area ID: 0.0.0.2
      Checksum: 0x3bee [correct]
      Instance ID: Base IPv4 Unicast Instance (0)
      Auth Type: Null (0)
      Auth Data (none): 0000000000000000
  ▾ LS Update Packet
      Number of LSAs: 9
    ▸ LSA-type 1 (Router-LSA), len 48
    ▸ LSA-type 1 (Router-LSA), len 72
    ▸ LSA-type 1 (Router-LSA), len 48
    ▸ LSA-type 2 (Network-LSA), len 32
    ▸ LSA-type 3 (Summary-LSA (IP network)), len 28
    ▸ LSA-type 3 (Summary-LSA (IP network)), len 28
    ▸ LSA-type 3 (Summary-LSA (IP network)), len 28
    ▸ LSA-type 4 (Summary-LSA (ASBR)), len 28
    ▸ LSA-type 5 (AS-External-LSA (ASBR)), len 36
```

```
0000  ca 0
0010  01 8
0020  01 0
0030  00 0
0040  22 0
0050  00 3
0060  02 8
0070  22 0
0080  00 4
0090  02 8
00a0  01 0
00b0  ff 0
00c0  00 0
00d0  01 0
00e0  00 0
00f0  00 0
0100  00 0
0110  00 0
0120  00 0
0130  00 0
0140  22 0
0150  00 1
0160  00 0
0170  00 0
0180  00 0
0190  00 6
```

e.

For each LSA type, the fields Link State ID and Advertising Router have specific meanings depending on the LSA, and are essential for building the network topology graph.

For the Router LSA (Type 1), the Link State ID represents the Router ID of the originating router, and the Advertising Router is also the Router ID of that router. These LSAs describe the router's interfaces and links. Important fields include the list of links, where each entry specifies the link type (point-to-point, transit, or stub), the Link ID (neighbor router or network), the Link Data (interface IP address or subnet mask), and the metric (cost). These elements are used to construct the nodes and edges of the topology graph.

For the Network LSA (Type 2), the Link State ID corresponds to the IP address of the Designated Router, and the Advertising Router is the Router ID of that DR. This LSA describes a multi-access network and includes the list of routers connected to that network. This allows the graph to represent the network as a central node connected to multiple routers.

For the Summary LSA (Type 3), the Link State ID represents the destination network being advertised, and the Advertising Router is the Router ID of the Area Border Router. This LSA contains the network mask and the metric, and is used to represent inter-area connectivity in the graph.

For the Summary ASBR LSA (Type 4), the Link State ID represents the Router ID of the ASBR, and the Advertising Router is the ABR that advertises it. This LSA includes the cost to reach the ASBR and is used to allow routers to reach external routing information.

For the AS-External LSA (Type 5), the Link State ID represents the external network being advertised, and the Advertising Router is the Router ID of the ASBR that injected the route. Important fields include the network mask, the metric, and the forwarding address. These LSAs represent routes external to the OSPF domain and are incorporated into the graph as external destinations.

f.

The latest Router LSA originated by R8 describes four links in area 2.

The first link is a point-to-point connection to router R6. In this entry, the ID field is 2.0.0.6, which corresponds to the Router ID of the neighboring router, and the Data field is 172.20.1.13, which is the IP address of R8's local interface on that link. This entry represents the direct serial connection between R8 and R6.

The second link is a stub network corresponding to the subnet of the serial link. In this case, the ID field is 172.20.1.12, which is the network address of the subnet, and the Data field is 255.255.255.252, which is the subnet mask. This entry represents the /30 network associated with the point-to-point connection.

The third link is a transit network. Here, the ID field is 172.20.1.10, which is the IP address of the Designated Router on that multi-access network, and the Data field is also 172.20.1.10, which is R8's own interface address on that network. This entry represents the broadcast segment shared with R9.

The fourth link is another stub network, corresponding to the LAN connected to R8. In this case, the ID field is 172.18.1.0, which is the network address of the LAN, and the Data field is 255.255.255.0, which is the subnet mask. This entry represents the directly connected local network of R8.

Together, these four entries describe both the router-to-router connectivity and the directly connected networks of R8, allowing the other routers to correctly build the topology graph and run the SPF calculation.

```
R8#show ip ospf database router 2.0.0.8

            OSPF Router with ID (2.0.0.8) (Process ID 2)

                Router Link States (Area 2)

  LS age: 933
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 2.0.0.8
  Advertising Router: 2.0.0.8
  LS Seq Number: 80000008
  Checksum: 0x6C4
  Length: 72
  Number of Links: 4

    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 2.0.0.6
     (Link Data) Router Interface address: 172.20.1.13
      Number of MTID metrics: 0
       TOS 0 Metrics: 647

    Link connected to: a Stub Network
     (Link ID) Network/subnet number: 172.20.1.12
     (Link Data) Network Mask: 255.255.255.252
      Number of MTID metrics: 0
       TOS 0 Metrics: 647

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 172.20.1.10
     (Link Data) Router Interface address: 172.20.1.10
      Number of MTID metrics: 0
       TOS 0 Metrics: 10

    Link connected to: a Stub Network
     (Link ID) Network/subnet number: 172.18.1.0
     (Link Data) Network Mask: 255.255.255.0
      Number of MTID metrics: 0
       TOS 0 Metrics: 10
```

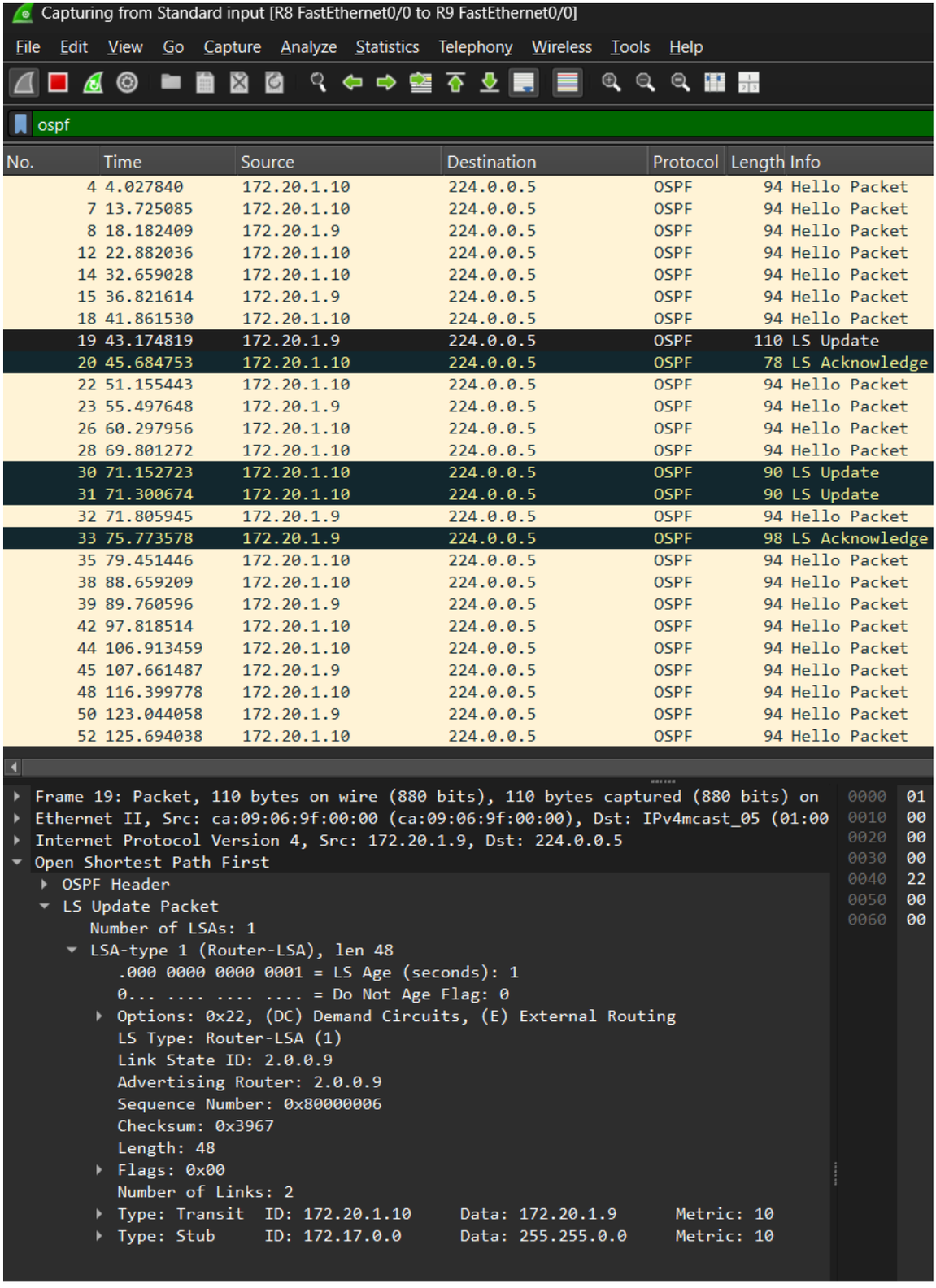
g.

After restarting R7, R8 sent LS Update messages to R9 containing multiple LSA types in order to update the LSDB and restore network convergence.

The following LSA types were identified:

The Type 1 (Router LSA), which describes the router's links and metrics, allowing the construction of the network topology graph. The Type 2 (Network LSA), which represents multi-access networks and the routers connected to them, generated by the Designated Router (DR). The Type 3 (Summary LSA), which advertises

networks from other areas, enabling inter-area routing. The Type 4 (ASBR Summary LSA), which provides reachability information to the ASBR. The Type 5 (AS External LSA), which carries routes external to the OSPF domain. These LSAs allow R9 to synchronize its LSDB and correctly compute routes after the topology change.



Capturing from Standard input [R8 FastEthernet0/0 to R9 FastEthernet0/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ospf

No.	Time	Source	Destination	Protocol	Length	Info
12	22.882036	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
14	32.659028	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
15	36.821614	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
18	41.861530	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
19	43.174819	172.20.1.9	224.0.0.5	OSPF	110	LS Update
20	45.684753	172.20.1.10	224.0.0.5	OSPF	78	LS Acknowledge
22	51.155443	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
23	55.497648	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
26	60.297956	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
28	69.801272	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
30	71.152723	172.20.1.10	224.0.0.5	OSPF	90	LS Update
31	71.300674	172.20.1.10	224.0.0.5	OSPF	90	LS Update
32	71.805945	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
33	75.773578	172.20.1.9	224.0.0.5	OSPF	98	LS Acknowledge
35	79.451446	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
38	88.659209	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
39	89.760596	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
42	97.818514	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
44	106.913459	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
45	107.661487	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
48	116.399778	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
50	123.044058	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
52	125.694038	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
55	134.887999	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet
57	140.270005	172.20.1.9	224.0.0.5	OSPF	94	Hello Packet
58	144.596672	172.20.1.10	224.0.0.5	OSPF	94	Hello Packet

```
▸ Frame 31: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on in
▸ Ethernet II, Src: ca:08:06:7f:00:00 (ca:08:06:7f:00:00), Dst: IPv4mcast_05 (01:00
▸ Internet Protocol Version 4, Src: 172.20.1.10, Dst: 224.0.0.5
▾ Open Shortest Path First
  ▸ OSPF Header
  ▾ LS Update Packet
      Number of LSAs: 1
    ▾ LSA-type 3 (Summary-LSA (IP network)), len 28
        .000 0000 0000 0010 = LS Age (seconds): 2
        0... .... .... .... = Do Not Age Flag: 0
      ▸ Options: 0x22, (DC) Demand Circuits, (E) External Routing
        LS Type: Summary-LSA (IP network) (3)
        Link State ID: 172.20.1.4
        Advertising Router: 2.0.0.6
        Sequence Number: 0x80000001
        Checksum: 0x213d
        Length: 28
        Netmask: 255.255.255.252
        TOS: 0
        Metric: 20
```

```
0000  01 00 5e 00 00
0010  00 4c 0a 3e 00
0020  00 05 02 04 00
0030  00 00 00 00 00
0040  22 03 ac 14 01
0050  00 1c ff ff ff
```

4.

a.

A Designated Router (DR) was elected on the broadcast interfaces FastEthernet0/0 and FastEthernet1/0, because OSPF treats these links as multi-access networks where DR/BDR election is required.

On FastEthernet0/0, a DR and BDR were elected (R8 as DR and R9 as BDR) since there are multiple OSPF routers on the same segment.

On FastEthernet1/0, R8 is also shown as DR, even though there are no neighbors, because it is still a broadcast network; however, no BDR is elected due to the absence of other routers.

No DR was elected on Serial4/0, because it is a point-to-point interface. In this type of network, OSPF does not perform DR/BDR election since communication occurs directly between two routers.

```
R8#sh ip ospf interface
Serial4/0 is up, line protocol is up
  Internet Address 172.20.1.13/30, Area 2, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.8, Network Type POINT_TO_POINT, Cost: 647
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           647       no          no           Base
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:00
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 4
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.0.0.6
  Suppress hello for 0 neighbor(s)
FastEthernet0/0 is up, line protocol is up
  Internet Address 172.20.1.10/30, Area 2, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.8, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.0.0.8, Interface address 172.20.1.10
  Backup Designated router (ID) 2.0.0.9, Interface address 172.20.1.9
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:08
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 4, maximum is 5
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.0.0.9  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
FastEthernet1/0 is up, line protocol is up
  Internet Address 172.18.1.1/24, Area 2, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.8, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.0.0.8, Interface address 172.18.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    No Hellos (Passive interface)
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

b.

A Network LSA (Type 2) is generated on broadcast networks where a DR is elected and there is at least one OSPF adjacency.

From the output, a Network LSA is present for the network 172.20.1.10/30, where R8 is the DR and both R8 and R9 are listed as attached routers. This confirms that the LSA is generated to describe the shared multi-access segment and the routers connected to it.

```
R8#sh ip ospf database network

            OSPF Router with ID (2.0.0.8) (Process ID 2)

                Net Link States (Area 2)

  Routing Bit Set on this LSA in topology Base with MTID 0
  LS age: 894
  Options: (No TOS-capability, DC)
  LS Type: Network Links
  Link State ID: 172.20.1.10 (address of Designated Router)
  Advertising Router: 2.0.0.8
  LS Seq Number: 80000005
  Checksum: 0x113D
  Length: 32
  Network Mask: /30
        Attached Router: 2.0.0.8
        Attached Router: 2.0.0.9
```

c.

The originator of a Network LSA is always the Designated Router (DR) of that network. This means that the router identified in the Advertising Router field of the Network LSA corresponds to the DR, which is responsible for describing the multi-access network and listing all attached routers.

5.

a.

On the multi-access network 192.168.100.0/24, the number of adjacencies depends on the role of each router.

R5 acts as the DR and has 3 adjacencies, since it forms full adjacencies with all other routers on the segment (R3, R4, and R6). R4 acts as the BDR and also has 3 adjacencies, for the same reason, maintaining full adjacency with all routers. R3 and R6 act as DROther routers and each has 2 adjacencies, as they only establish adjacencies with the DR (R5) and the BDR (R4), but not with each other.

This behavior reduces the number of adjacencies required in a broadcast network while maintaining full connectivity through the DR and BDR.

```
R6#show ip ospf interface f1/0
FastEthernet1/0 is up, line protocol is up
  Internet Address 192.168.100.2/24, Area 0, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.6, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 2.0.0.5, Interface address 192.168.100.254
  Backup Designated router (ID) 2.0.0.4, Interface address 192.168.100.253
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:04
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 8
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 2
    Adjacent with neighbor 2.0.0.4  (Backup Designated Router)
    Adjacent with neighbor 2.0.0.5  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

```
R5#show ip ospf interface f0/0
FastEthernet0/0 is up, line protocol is up
  Internet Address 192.168.100.254/24, Area 0, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.5, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.0.0.5, Interface address 192.168.100.254
  Backup Designated router (ID) 2.0.0.4, Interface address 192.168.100.253
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:07
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 8
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 2.0.0.3
    Adjacent with neighbor 2.0.0.4  (Backup Designated Router)
    Adjacent with neighbor 2.0.0.6
  Suppress hello for 0 neighbor(s)
```

```
R4#show ip ospf interface f3/0
FastEthernet3/0 is up, line protocol is up
  Internet Address 192.168.100.253/24, Area 0, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.4, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.0.0.5, Interface address 192.168.100.254
  Backup Designated router (ID) 2.0.0.4, Interface address 192.168.100.253
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:08
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 1
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 3
    Adjacent with neighbor 2.0.0.3
    Adjacent with neighbor 2.0.0.5  (Designated Router)
    Adjacent with neighbor 2.0.0.6
  Suppress hello for 0 neighbor(s)
```

```
R3#show ip ospf interface f2/0
FastEthernet2/0 is up, line protocol is up
  Internet Address 192.168.100.1/24, Area 0, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.3, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 2.0.0.5, Interface address 192.168.100.254
  Backup Designated router (ID) 2.0.0.4, Interface address 192.168.100.253
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 4
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 3, Adjacent neighbor count is 2
    Adjacent with neighbor 2.0.0.4  (Backup Designated Router)
    Adjacent with neighbor 2.0.0.5  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

b.

The Designated Router (DR) is used to reduce the number of OSPF adjacencies and optimize LSA flooding in multi-access networks, acting as a central point for exchanging routing information.

6.

a.

```
PC1> trace 172.16.1.2
trace to 172.16.1.2, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   9.014 ms  8.638 ms  9.896 ms
 2   192.168.1.130   30.132 ms  19.864 ms  20.073 ms
 3   192.168.50.2   30.026 ms  39.706 ms  41.087 ms
 4   192.168.100.2   73.873 ms  52.107 ms  52.533 ms
 5   172.20.1.2   72.112 ms  72.164 ms  61.177 ms
 6   *172.16.1.2   73.504 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> trace 172.16.1.2
trace to 172.16.1.2, 8 hops max, press Ctrl+C to stop
 1   192.168.1.1   9.211 ms  9.709 ms  9.098 ms
 2   *192.168.1.1   10.499 ms (ICMP type:3, code:1, Destination host unreachable)
```

```
R6#sh ip ospf database

            OSPF Router with ID (2.0.0.6) (Process ID 2)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
2.0.0.2         2.0.0.2         293         0x80000002 0x007371 1
2.0.0.3         2.0.0.3         292         0x80000003 0x002FFB 2
2.0.0.4         2.0.0.4         296         0x80000002 0x00730E 1
2.0.0.5         2.0.0.5         301         0x80000002 0x00710D 1
2.0.0.6         2.0.0.6         301         0x80000002 0x008EE9 1

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
192.168.50.2    2.0.0.3         296         0x80000001 0x00EBA2
192.168.100.2   2.0.0.6         291         0x80000003 0x00E459

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
172.17.0.0      2.0.0.6         288         0x80000001 0x00E7F1
172.18.1.0      2.0.0.6         355         0x80000001 0x006C75
172.20.1.8      2.0.0.6         355         0x80000001 0x00F1E8
172.20.1.12     2.0.0.6         355         0x80000001 0x00657B

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
2.0.0.6         2.0.0.6         208         0x80000003 0x001020 0
2.0.0.7         2.0.0.7         295         0x80000003 0x0059B5 3
2.0.0.9         2.0.0.9         299         0x80000002 0x002C57 1

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
172.20.1.2      2.0.0.7         304         0x80000001 0x003B24
172.20.1.6      2.0.0.9         303         0x80000001 0x00292D
```

```
                Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
172.17.0.0      2.0.0.6         288         0x80000001 0x00E7F1
172.18.1.0      2.0.0.6         355         0x80000001 0x006C75
172.20.1.8      2.0.0.6         355         0x80000001 0x00F1E8
172.20.1.12     2.0.0.6         355         0x80000001 0x00657B
192.168.50.0    2.0.0.6         288         0x80000001 0x004243
192.168.100.0   2.0.0.6         355         0x80000001 0x00B5A7

                Summary ASB Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
2.0.0.2         2.0.0.6         278         0x80000001 0x00DF3C

                Router Link States (Area 2)

Link ID         ADV Router      Age         Seq#       Checksum Link c
2.0.0.6         2.0.0.6         364         0x80000001 0x003527 2
2.0.0.8         2.0.0.8         302         0x80000003 0x00EDE2 4
2.0.0.9         2.0.0.9         299         0x80000002 0x00376E 2

                Net Link States (Area 2)

Link ID         ADV Router      Age         Seq#       Checksum
172.20.1.9      2.0.0.9         303         0x80000001 0x001939

                Summary Net Link States (Area 2)

Link ID         ADV Router      Age         Seq#       Checksum
192.168.50.0    2.0.0.6         288         0x80000001 0x004243
192.168.100.0   2.0.0.6         355         0x80000001 0x00B5A7

                Summary ASB Link States (Area 2)

Link ID         ADV Router      Age         Seq#       Checksum
2.0.0.2         2.0.0.6         278         0x80000001 0x00DF3C

                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum Tag
192.168.1.0     2.0.0.2         331         0x80000001 0x0030B1 0
```

b.

The path change is not achieved after disconnecting the link between R6 and R7 because OSPF is unable to compute a valid alternative route to the destination.

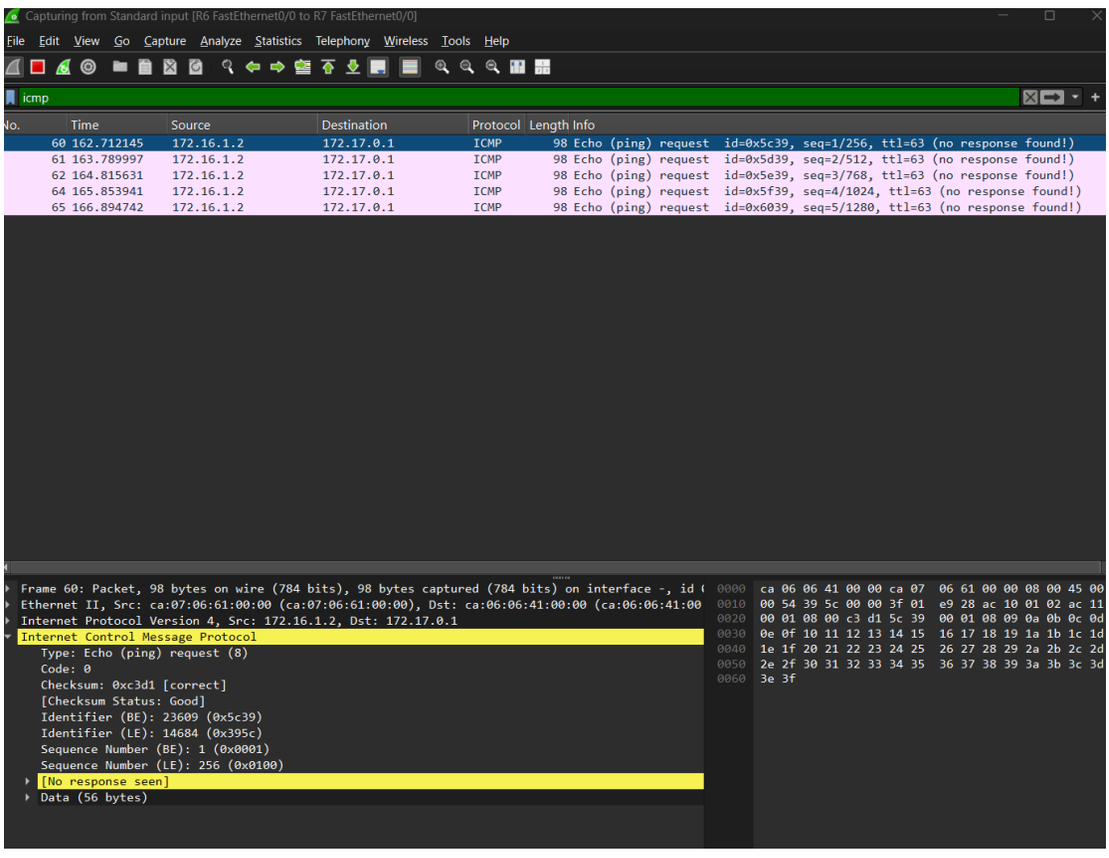
From the OSPF database in R6, although there are LSAs describing routers R8 and R9 in Area 2, there is no complete and consistent path that ensures reachability to the destination network after the link failure. As a result, the SPF algorithm does not install a new route.

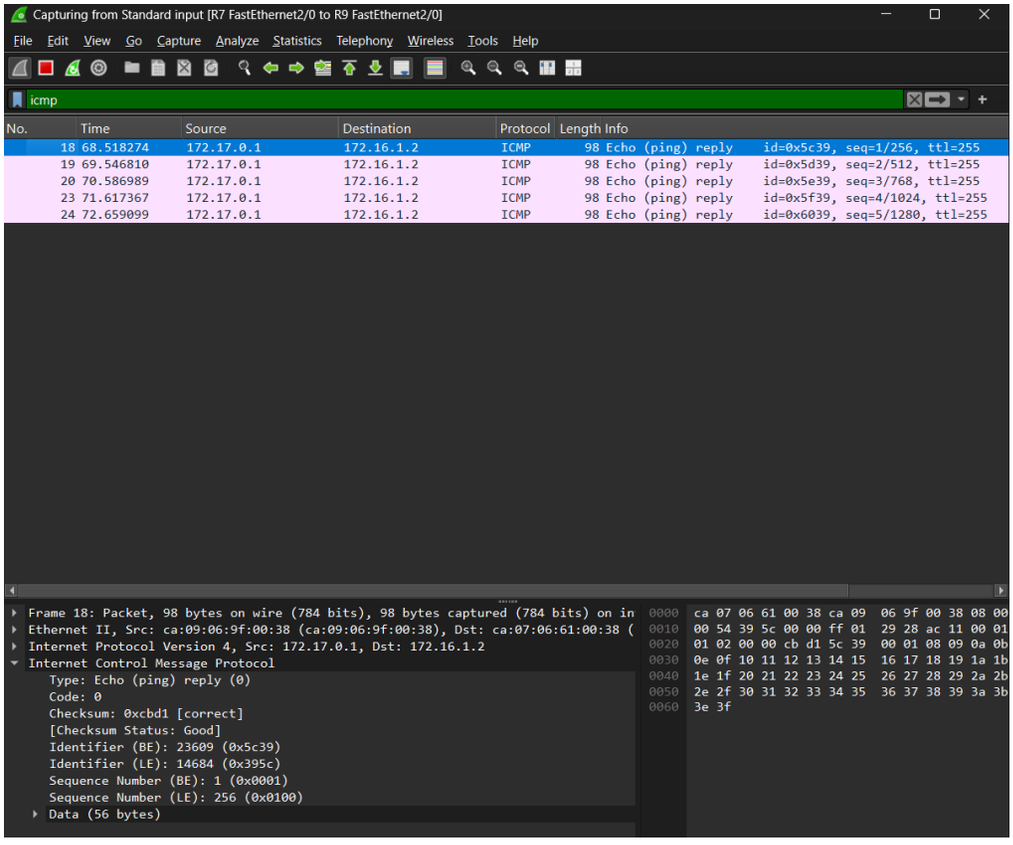
This behavior is confirmed by the traceroute, where the packet fails early with a “destination unreachable” message, indicating that no valid route exists in the routing table.

Therefore, the traffic cannot be redirected through R8 and R9 because the network topology, as described in the LSDB, does not provide a valid alternative path after the failure.

7.

a.





From the captures on R7, the ICMP echo request (from 172.16.1.2 to 172.17.0.1) is observed on the interface connected to R6, indicating that the request follows the path through R6 - R7 - R9 towards the destination.

The ICMP echo reply (from 172.17.0.1 to 172.16.1.2) is observed on the interface connected to R9, indicating that the return traffic follows the path R9 - R7 - R6 back to the source.

This behavior occurs because OSPF performs routing decisions independently on each router based on its own routing table. As a result, the forward and return paths may differ depending on the calculated shortest path from each router's perspective.

b.

```
R9#sh ip ospf interface f2/0
FastEthernet2/0 is up, line protocol is up
  Internet Address 172.20.1.6/30, Area 1, Attached via Network Statement
  Process ID 2, Router ID 2.0.0.9, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
```

R9(config)#interface f2/0

R9(config-if)#ip ospf cost

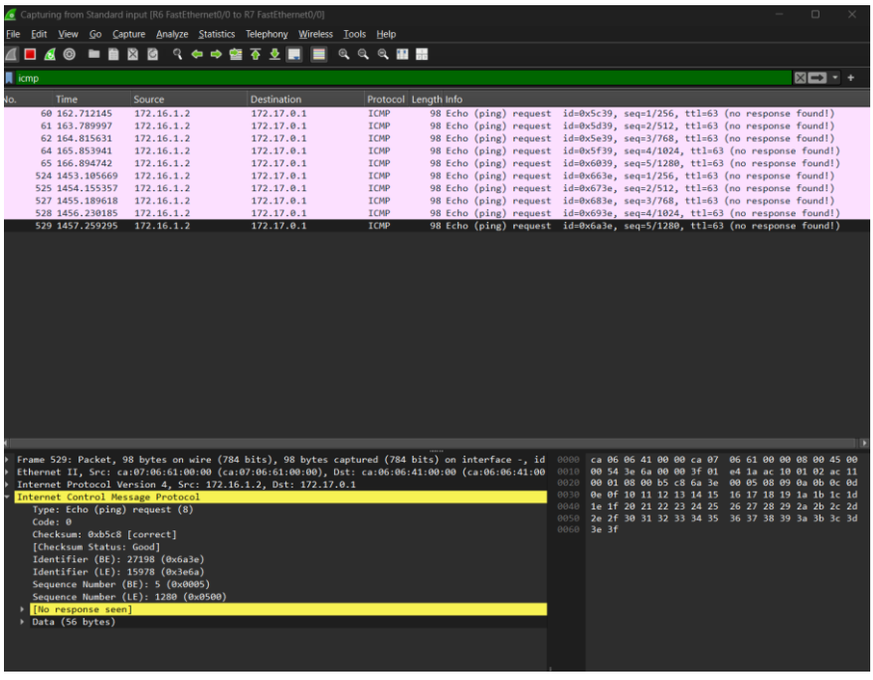
R9(config-if)#ip ospf cost 100

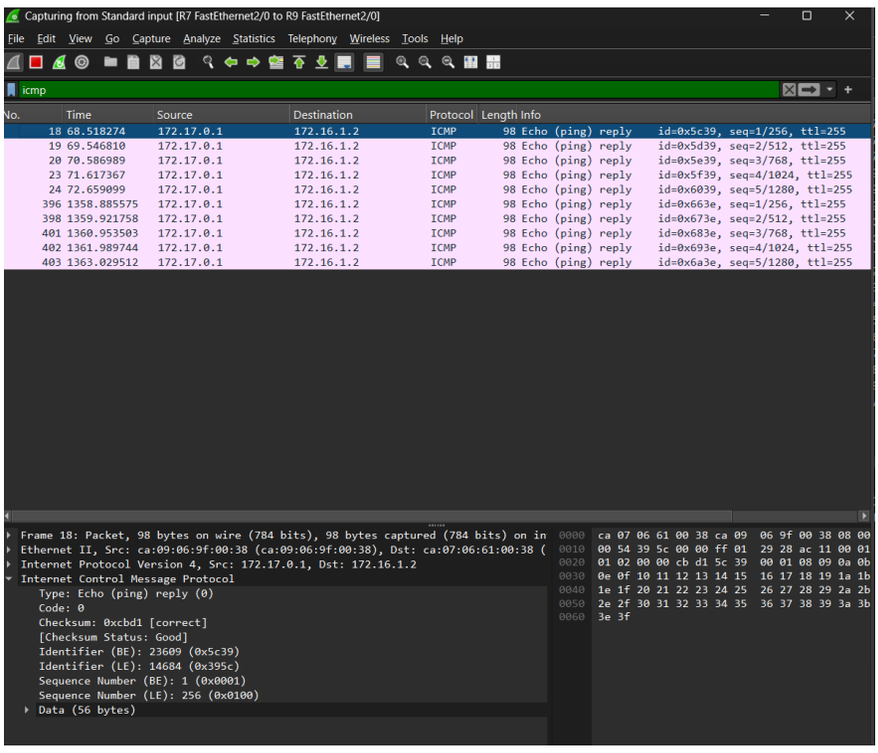
R9(config-if)#do wr

It was not possible to make the echo reply follow the exact inverse path of the echo request only by changing the OSPF cost of interface 172.20.1.6 on R9.

Although the interface cost was increased, OSPF still selected the direct path through R9 → R7 for the reply, because that path remained valid and preferred according to the topology and SPF calculation. Therefore, changing the cost of a single interface was not enough to force the reverse path to match the forward path.

When the link between R9 and R7 was disconnected, the direct path was no longer available, and the reply was forced to use the alternative route. In that case, both the echo request and the echo reply were observed through the same remaining path.





Capturing from Standard input [R6 FastEthernet0/0 to R7 FastEthernet0/0]

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
60	162.712145	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x5c39, seq=1/256, ttl=63 (no response found!)
61	163.789997	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x5d39, seq=2/512, ttl=63 (no response found!)
62	164.815631	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x5e39, seq=3/768, ttl=63 (no response found!)
64	165.853941	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x5f39, seq=4/1024, ttl=63 (no response found!)
65	166.894742	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x6039, seq=5/1280, ttl=63 (no response found!)
524	1453.105669	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x663e, seq=1/256, ttl=63 (no response found!)
525	1454.155357	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x673e, seq=2/512, ttl=63 (no response found!)
527	1455.189618	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x683e, seq=3/768, ttl=63 (no response found!)
528	1456.230185	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x693e, seq=4/1024, ttl=63 (no response found!)
529	1457.259295	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x6a3e, seq=5/1280, ttl=63 (no response found!)
596	1638.799713	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x203f, seq=1/256, ttl=63 (reply in 597)
597	1638.840261	172.17.0.1	172.16.1.2	ICMP	98	Echo (ping) reply id=0x203f, seq=1/256, ttl=253 (request in 596)
599	1639.853666	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x213f, seq=2/512, ttl=63 (reply in 600)
600	1639.887562	172.17.0.1	172.16.1.2	ICMP	98	Echo (ping) reply id=0x213f, seq=2/512, ttl=253 (request in 599)
601	1640.901371	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x223f, seq=3/768, ttl=63 (reply in 602)
602	1640.947118	172.17.0.1	172.16.1.2	ICMP	98	Echo (ping) reply id=0x223f, seq=3/768, ttl=253 (request in 601)
604	1641.962559	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x233f, seq=4/1024, ttl=63 (reply in 605)
605	1641.988063	172.17.0.1	172.16.1.2	ICMP	98	Echo (ping) reply id=0x233f, seq=4/1024, ttl=253 (request in 604)
606	1643.005335	172.16.1.2	172.17.0.1	ICMP	98	Echo (ping) request id=0x243f, seq=5/1280, ttl=63 (reply in 607)
607	1643.029958	172.17.0.1	172.16.1.2	ICMP	98	Echo (ping) reply id=0x243f, seq=5/1280, ttl=253 (request in 606)

```
▸ Frame 529: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id
▸ Ethernet II, Src: ca:07:06:61:00:00 (ca:07:06:61:00:00), Dst: ca:06:06:41:00:00 (ca:06:06:41:00
▸ Internet Protocol Version 4, Src: 172.16.1.2, Dst: 172.17.0.1
▾ Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0xb5c8 [correct]
    [Checksum Status: Good]
    Identifier (BE): 27198 (0x6a3e)
    Identifier (LE): 15978 (0x3e6a)
    Sequence Number (BE): 5 (0x0005)
    Sequence Number (LE): 1280 (0x0500)
  ▸ [No response seen]
  ▸ Data (56 bytes)
```

```
0000  ca 06 06 41 00 00 ca 07  06 61 00 00 08 00 45 00
0010  00 54 3e 6a 00 00 3f 01  e4 1a ac 10 01 02 ac 11
0020  00 01 08 00 b5 c8 6a 3e  00 05 08 09 0a 0b 0c 0d
0030  0e 0f 10 11 12 13 14 15  16 17 18 19 1a 1b 1c 1d
0040  1e 1f 20 21 22 23 24 25  26 27 28 29 2a 2b 2c 2d
0050  2e 2f 30 31 32 33 34 35  36 37 38 39 3a 3b 3c 3d
0060  3e 3f
```